Equilux

Pay-gap reports anyone can check.
No salary revealed.

Wave 3: the report now covers any company up to 512 people, proven 16 at a time on Midnight. Employees check their own record with no wallet. A 32-person company is on Midnight's public testnet: 58 transactions, all three reports proven.

Midnight Buildathon · Wave 3 | equilux-lac.vercel.app | github.com/AustinChris1/Equilux

— THE LAW

From 7 June 2027, EU employers must publish their gender pay gap.

250+

WORKERS: EVERY YEAR

150–249

EVERY THREE YEARS · 100–149 FROM 2031

~53,000

EU BUSINESSES WITH 250+ STAFF (EUROSTAT)

> Today the report is **self-declared**. Nobody outside can check it without seeing everyone's salary. A report can leave people out or bend a figure, and nobody can tell.

What the judges asked, and what we built.

THE QUESTION	WAVE 2	WAVE 3
Does it work at the size the law means?	~~16 people per report~~	**Up to 512**, proven in batches of 16, running totals sealed between batches. 250 tested.
Who pays for it?	~~not addressed~~	Measured cost structure, desk research, a pilot page, and five notes to EU people who work on this report
Less wallet friction	~~every step needs a wallet~~	Employees **check their own record with no wallet, no fee** (`/check`)
What's beyond Article 9?	~~two reports~~	A third report: **the gap inside equal job levels**, with each level's mix always shown
Can a newcomer follow it?	~~a README~~	Plain-language `/docs`, a 90-second Wave 3 film, a guided app

HOW IT WORKS

Five people, five keys, one proof.

01

Payroll provider seals the payroll

Every row becomes a sealed code on Midnight. Salaries stay with payroll.

02

Works council checks it

Opens each sealed row against the real payroll and approves.

03

Each employee confirms

A record must match its row. An inflated salary matches nothing.

04

Employer proves the figures

All seven Article 9 figures, checked inside the circuits, 16 people at a time.

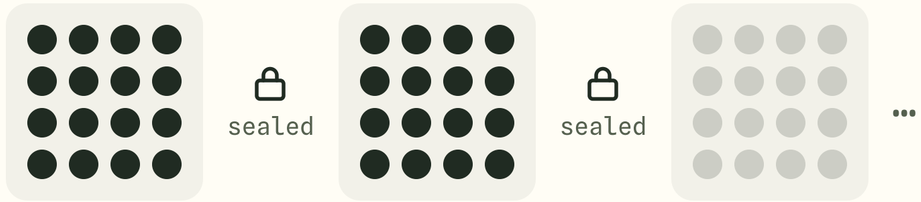
05

Anyone verifies

Read the report from the chain. Employees check their own record.

Salaries, genders, job levels and secrets are private inputs on each person's device. Only sealed codes, anonymous marks and the proven figures reach the chain.

Any company up to 512 people. Sixteen at a time.



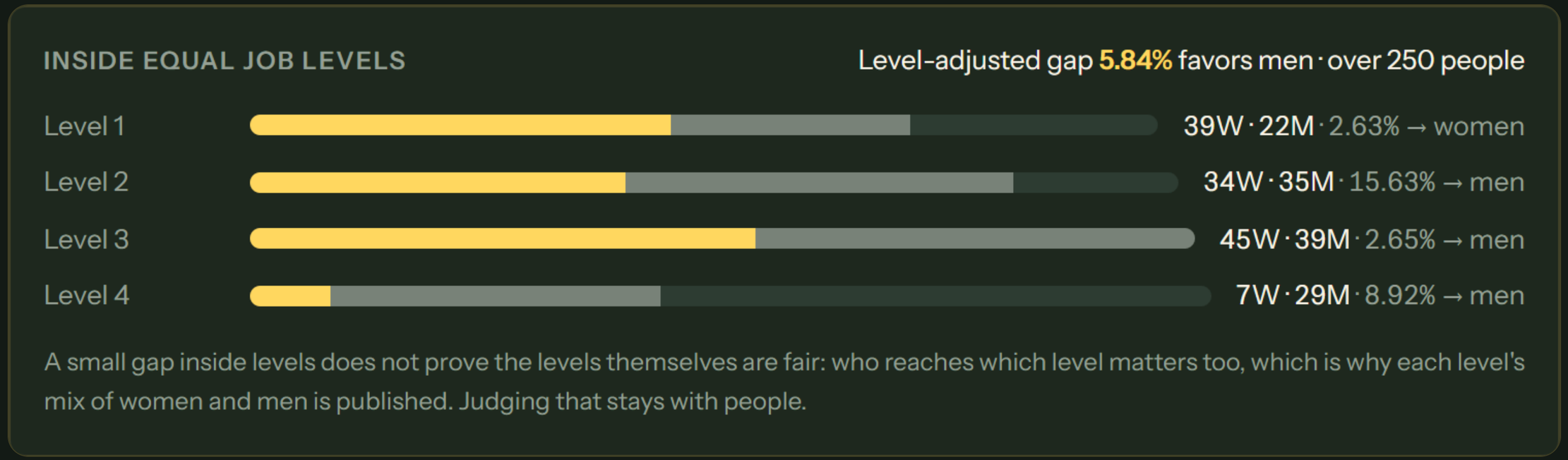
Each report is **begin** → **batch** × ⌈**n/16**⌉ → **finish**. Between batches the running totals are **sealed**: the chain holds only `hash(totals, claim, salt)`, so no batch reveals what its 16 people earn. Each record is counted once per report.

COMPANY	BATCHES	TRANSACTIONS
32 people `on preprod`	2	**58** (measured)
150 people	10	216
250 people `tested`	16	346
512 people	32	688

n + 5·⌈n/16⌉ + 16 transactions; n of them are the employees' own. 73 tests, including a 250-person company through the compiled circuits.

NEW IN WAVE 3 · BEYOND ARTICLE 9

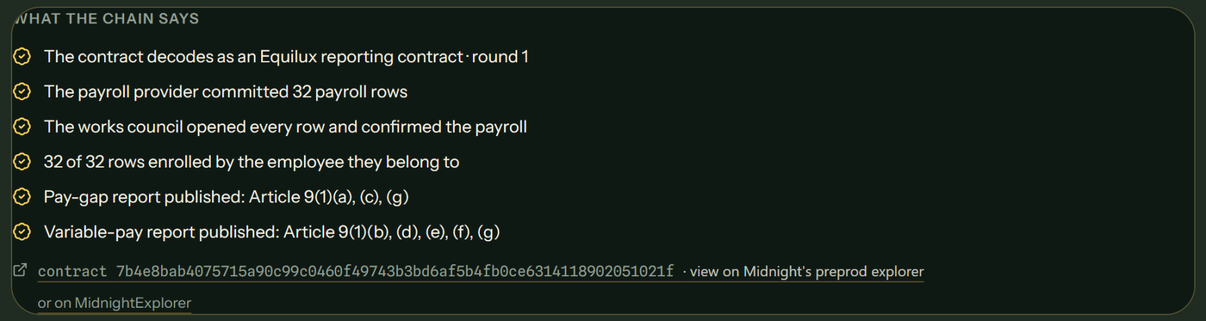
The gap inside equal job levels.



> Each level's mix of women and men is **always** published. A small gap inside levels does not prove the levels themselves are fair: who reaches which level matters too, and judging that stays with people.

EVIDENCE · MIDNIGHT PREPROD

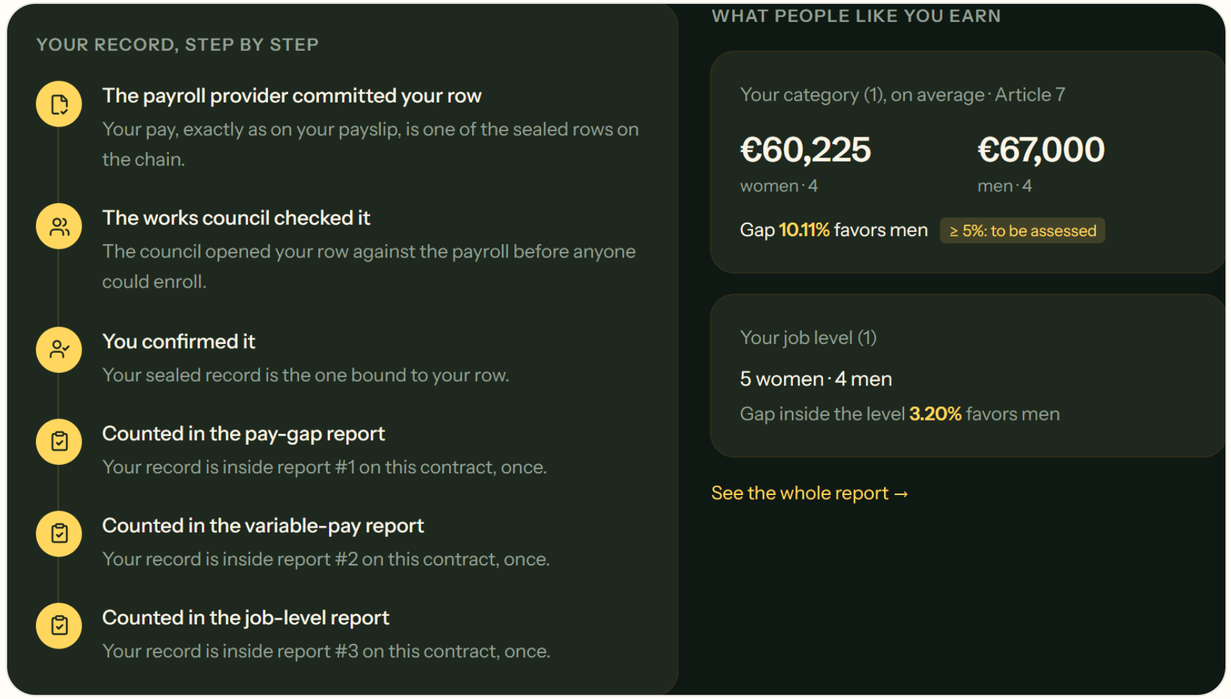
A 32-person company, proven on the public testnet.



- **58 transactions**, blocks 3,223,740 → 3,224,162
- Every batched step ran twice (2 batches of 16)
- Mean gap **9.50%** · median **3.40%** · level-adjusted **9.05%**
- 32 of 32 rows checked by the council and confirmed by their employee
- **0 salaries** on chain

Contract 7b4e8bab…2051021f · every tx in `contract/preprod/deployment-v4.json`

Employees check their own record from a payslip.



- My row was committed, the council checked it, I confirmed it, I was counted **once in each report**
- What women and men in **my group** earn (Article 7)
- **No wallet, no fee.** Details stay in the page; the sealed codes are recomputed by the compiled contract
- Change the salary by €100: **no row matches**

Where Equilux stands, stated plainly.

Live product

The whole report runs in the browser, nothing to install. `/verify` and `/check` read the real chain. Docs at `/docs`.

Users: early

No company uses or pays for it yet. Free pilots are open, and visits are counted anonymously so the numbers we report are real, even small.

Going to the buyers

Five notes to EU people who already work on this report: works-council trainers, an equal-pay consultancy, a pay-transparency software maker, a fair-pay certifier.

> Every reply goes into the submission as it is, including "no", and including silence.

The employer pays, per employee, through the tools it already uses.

1 · Software partners

Pay-transparency and payroll software already make the report. Equilux adds **proof** it covers the whole payroll; they resell it.

2 · Legal risk

In a dispute the employer must prove it did not discriminate (Art. 18). A proven report is evidence a spreadsheet is not.

3 · Councils & certifiers

Works councils ask for a report they can check; fair-pay certifiers back their certificate with proof. They create demand.

Why **per employee per year**: the work grows with headcount, almost one transaction per person. No price is set: mainnet fees are unknown and no buyer has been asked yet. Pay-equity tools (Syndio, Trusaic, Figures) analyse pay; none makes the report checkable without showing salaries, so Equilux sits beside them.

Taking it toward the market.

Learned

Cost grows with headcount. Wallets are the adoption barrier, so we split checking (walletless now) from enrolling. Midnight's preprod indexer closed mid-Buildathon: moved to Blockfrost in a week. A 14-circuit contract doesn't deploy in one block: split it.

Challenges

Reaching HR buyers with no network yet. Walletless enrollment needs an always-on synced wallet that pays fees (a cold preprod sync took 2.5 hours). Mainnet fees unknown.

How Midnight can help

Fee sponsorship, so employers pay every employee's fee. An introduction to one HR or payroll partner. A server-side wallet and hosted proving a relayer can rely on.

Equilux

Try it. Check it.
Run a pilot with us.

The app	equilux-lac.vercel.app/app
A real report	equilux-lac.vercel.app/verify
Your own record	equilux-lac.vercel.app/check
Docs	equilux-lac.vercel.app/docs
Free pilots	equilux-lac.vercel.app/#pilot
Source	github.com/AustinChris1/Equilux